

Change Management Procedure for Data Assets

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APPROVAL SIGNATURES

Role	Name	Signature	Date
Chief Data Officer	Dr. Khalid Al-Otaibi	[Signed]	June 15, 2024
IT Operations Director	Mohammed Al-Zahrani	[Signed]	June 14, 2024
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TABLE OF CONTENTS

1. PURPOSE & SCOPE	4
2. CHANGE CATEGORIES & SLA	6
3. THE CHANGE LIFECYCLE	8
4. IMPACT ASSESSMENT MATRIX	12
5. APPROVAL WORKFLOWS	14
6. ROLLBACK PROCEDURES	16
7. COMMUNICATION REQUIREMENTS	18
8. ROLES AND RESPONSIBILITIES	19
APPENDIX A: CHANGE REQUEST TEMPLATE	20

1. PURPOSE & SCOPE

1.1 Purpose

This procedure defines the standardized method for handling all changes to Data Assets at Company A. The objective is to ensure minimal disruption to services while maintaining data integrity, quality, and regulatory compliance.

1.2 Scope

In Scope

Change Type	Examples	Governance Level
Database Schema Changes (DDL)	Adding columns, modifying constraints	High
Data Model Changes	Entity modifications, relationship changes	High
Data Policy Updates	Retention, access, classification changes	Medium
Metadata Definitions	Business glossary, technical descriptions	Low
Reference Data Updates	Adding codes, modifying lookups	Medium
ETL/Integration Changes	Job modifications, mapping changes	Medium
Report Changes	Dashboard modifications, new metrics	Low

Out of Scope

- Infrastructure changes (handled by IT Change Management)
- Application code changes (handled by Software Development)
- Network/Security changes (handled by Security Operations)

1.3 NDMO Alignment

NDMO Control	Requirement	Implementation
DG-03	Change management for data assets	This procedure
DG-04	Impact assessment requirements	Section 4
DG-05	Approval workflows	Section 5

1.4 Definitions

Term	Definition
Change Request (CR)	Formal submission for a data asset change
Change Advisory Board (CAB)	Committee reviewing major/emergency changes
Data Governance Committee (DGC)	Executive body for policy changes
Rollback	Reverting a change to previous state
Change Window	Approved time period for implementing changes

2. CHANGE CATEGORIES & SLA

2.1 Category Definitions

Standard Change

Attribute	Value
Definition	Pre-approved, routine, low-risk changes
Examples	Adding descriptions, updating non-critical metadat
Risk Level	Low
Approval	Data Steward
SLA	2 business days
Testing Required	Minimal

Minor Change

Attribute	Value
Definition	Low-risk changes requiring assessment
Examples	Adding non-mandatory columns, new reference codes
Risk Level	Low-Medium
Approval	Data Owner
SLA	5 business days
Testing Required	Standard

Major Change

Attribute	Value
Definition	High-impact changes affecting multiple systems
Examples	Modifying primary keys, table restructuring
Risk Level	High
Approval	CAB + DGC
SLA	14 business days
Testing Required	Full regression

Emergency Change

Attribute	Value
Definition	Critical changes to fix outages or security issues
Examples	Fixing data corruption, security vulnerabilities
Risk Level	Critical
Approval	Emergency CAB
SLA	4 hours
Testing Required	As feasible

2.2 SLA Summary

Category	Submission to Approval	Approval to Implementation	Total SLA
Standard	1 day	1 day	2 days
Minor	3 days	2 days	5 days
Major	10 days	4 days	14 days

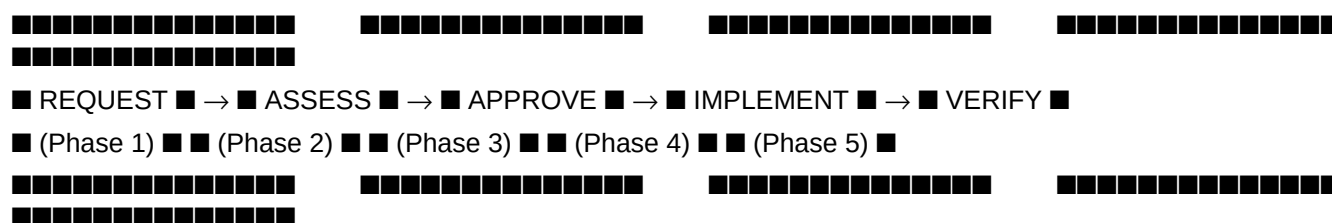
Emergency	2 hours	2 hours	4 hours
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2.3 Escalation Path

Delay Threshold	Escalation To	Action
SLA + 1 day	Data Steward Manager	Email reminder
SLA + 3 days	Data Owner	Meeting scheduled
SLA + 5 days	CDO	Executive escalation

3. THE CHANGE LIFECYCLE

3.1 Lifecycle Overview



3.2 Phase 1: Request (CR Creation)

3.2.1 Submission Process

Step	Action	Tool
1	Log into ServiceNow	ITSM Portal
2	Select "Data Change Request"	Module
3	Complete required fields	Form
4	Attach supporting documentation	Attachments
5	Submit for assessment	Workflow

3.2.2 Required Fields

Field	Description	Mandatory
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Change Title	Brief description	Yes
Business Justification	Why is this change needed?	Yes
Data Asset(s) Affected	Tables, columns, systems	Yes
Requested Category	Standard/Minor/Major	Yes
Risk Assessment	Initial risk evaluation	Yes
Rollback Plan	How to reverse the change	Yes
Implementation Window	Preferred date/time	Yes
Testing Requirements	What testing is planned	Yes

3.3 Phase 2: Assessment

3.3.1 Technical Assessment

Assessment Area	Tool	Output
Impact Analysis	Informatica EDC Lineage	List of impacted assets
Data Quality Impact	Informatica IDQ	DQ rule changes needed
Performance Impact	Database explain plans	Performance estimates
Security Impact	Classification review	Access changes needed

3.3.2 Impact Analysis Questions

Question	Data Source
Who consumes this data?	EDC lineage graph
What reports will be affected?	Report catalog
Are there downstream APIs?	API registry
What ETL jobs use this data?	Job dependencies
Are there data sharing agreements?	DSA register

3.3.3 Assessment Output

The assessment produces a formal Impact Report containing:

- List of all impacted systems
- List of impacted data consumers
- Required notifications
- Testing recommendations
- Risk rating (Low/Medium/High/Critical)

3.4 Phase 3: Approval

3.4.1 Approval Workflow

Category	Approver 1	Approver 2	Approver 3
Standard	Data Steward	-	-
Minor	Data Steward	Data Owner	-
Major	Data Steward	Data Owner	CAB + DGC
Emergency	On-call Manager	Emergency CAB	-

3.4.2 CAB Meeting Schedule

Meeting	Frequency	Time	Chair
Regular CAB	Weekly	Wednesday 10:00	IT Ops Director
Emergency CAB	As needed	Within 2 hours	On-call Manager

3.5 Phase 4: Implementation

3.5.1 Environment Promotion

Environment	Purpose	Approval Required
DEV	Initial development	None
TEST	Integration testing	Tech Lead
UAT	User acceptance testing	Data Owner
PROD	Production deployment	Change Approvers

3.5.2 Change Windows

Category	Allowed Window	Blackout Periods
Standard	Business hours	None
Minor	Business hours	None
Major	Thursday 00:00-04:00	Month-end, Year-end
Emergency	Anytime	None

3.5.3 Implementation Checklist

Step	Action	Verification
1	Create backup/snapshot	Confirm backup exists
2	Execute change script	Script completes
3	Verify data integrity	Row counts match
4	Run DQ checks	All rules pass
5	Test dependent systems	Applications functional
6	Update metadata	Catalog reflects change

3.6 Phase 5: Verification

3.6.1 Post-Implementation Review

Check	Method	Timing
Data Quality	Automated DQ rules	Immediate
Application Function	Smoke tests	Within 1 hour
User Acceptance	User confirmation	Within 24 hours
Performance	Monitoring dashboards	Within 24 hours

3.6.2 Closure Requirements

Requirement	Verification
All tests passed	Test results attached
No critical issues	Incident log reviewed
User confirmation	Email confirmation
Documentation updated	Metadata current

4. IMPACT ASSESSMENT MATRIX

4.1 Impact Dimensions

Dimension	Low Impact	Medium Impact	High Impact
Data Volume	< 10,000 rows	10,000 - 1M rows	> 1M rows
Classification	PUBLIC	INTERNAL	CONFIDENTIAL/SECRET
Integration	No external APIs	Internal APIs	External APIs
Downtime	None	< 30 minutes	> 30 minutes
Users Affected	< 10 users	10-100 users	> 100 users
Reports	No reports	Internal reports	Executive/External reports

4.2 Risk Calculation

Total Impact Score	Risk Rating	Category Minimum
1-6	Low	Standard
7-12	Medium	Minor
13-18	High	Major
19+	Critical	Major + Executive

4.3 Special Considerations

Scenario	Automatic Escalation
Patient data affected	Major minimum
Financial data affected	Major minimum
Regulatory impact	DGC approval required
External reporting impact	Executive notification

5. APPROVAL WORKFLOWS

5.1 Standard Change Workflow

Requester → Data Steward → Auto-approve → Implement → Verify → Close

5.2 Minor Change Workflow

Requester → Data Steward → Data Owner → Implement → Verify → Close

5.3 Major Change Workflow

Requester → Data Steward → Data Owner → CAB Review → DGC Approval → Schedule → Implement → Verify → PIR → Close

5.4 Emergency Change Workflow

Requester → On-call Manager → Emergency CAB → Immediate Implement → Verify → Retrospective Review → Close

6. ROLLBACK PROCEDURES

6.1 Rollback Requirements

Mandatory Rule: No Major or Emergency Change Request can be approved without a tested rollback procedure.

6.2 Rollback Methods by Change Type

Change Type	Rollback Method	RTO
DDL Changes	FLASHBACK TABLE	15 minutes
Data Updates	Point-in-time restore	30 minutes
ETL Changes	Previous job version	15 minutes
Metadata Changes	Repository version revert	5 minutes
Reference Data	Previous code version	10 minutes

6.3 Rollback Decision

Condition	Action
Critical application failure	Immediate rollback
Data quality below threshold	Rollback within 2 hours
User rejection in UAT	Rollback before go-live
Performance degradation > 20%	Rollback and investigate

6.4 Rollback Execution

- Step 1: Incident declared
- ↓
- Step 2: Rollback approved by Change Owner
- ↓
- Step 3: Execute rollback script
- ↓
- Step 4: Verify system restored
- ↓
- Step 5: Notify stakeholders
- ↓
- Step 6: Root cause analysis scheduled

7. COMMUNICATION REQUIREMENTS

7.1 Notification Matrix

Change Category	Pre-Implementation	Post-Implementation	Failure
Standard	None	None	Steward
Minor	24 hours (email)	Confirmation email	Owner + Steward
Major	1 week (formal)	Formal announcement	All stakeholders
Emergency	ASAP (all channels)	Within 4 hours	Executive notification

7.2 Communication Channels

Channel	Use Case
Email	Standard notifications
ServiceNow	Change tracking
MS Teams	Real-time coordination
SMS	Emergency alerts

8. ROLES AND RESPONSIBILITIES

8.1 RACI Matrix

Activity	Requester	Steward	Owner	CAB	DGC
Submit CR	R	I	I	I	I
Assessment	C	R	A	C	I
Approval	I	R	A	A/R	A/R
Implementation	R	C	A	I	I
Verification	R	R	A	I	I
Rollback decision	C	R	A	A	I

R = Responsible, A = Accountable, C = Consulted, I = Informed

APPENDIX A: CHANGE REQUEST TEMPLATE

Data Change Request Form

Field	Value
CR Number	DCR-2024-_____
Requester	_____
Date Submitted	_____
Category	☐ Standard ☐ Minor ☐ Major ☐ Emergency

Change Description

Section	Content
Title	_____
Business Justification	_____
Technical Description	_____
Data Assets Affected	_____
Systems Impacted	_____

Risk Assessment

Dimension	Score (1-3)
Data Volume	—
Classification	—
Integration	—
Downtime	—
Total	—

Implementation Plan

Step	Description	Duration
1		
2		
3		

Rollback Plan

Step	Description	Duration
1		
2		

Approvals

Role	Name	Signature	Date
Data Steward			
Data Owner			
CAB Chair			

Document End

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